



Sam Rahnemayan

MSc Robotics Candidate @ EPFL



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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

EPFL MSc Robotics candidate specializing in rehabilitation robotics, biomechanical systems, embedded systems, autonomous control, and human-machine interaction. Experienced in translating research concepts into deployable robotic and medical technology systems.

Education

École Polytechnique Fédérale de Lausanne (EPFL)

Master in Robotics

Minor in Management, Technology and Entrepreneurship

Master Thesis: *Neuromuscular Adaptation to Exoskeleton Assistance*

Lausanne, Switzerland

Feb 2023 – Apr 2026

École Polytechnique Fédérale de Lausanne (EPFL)

Bachelor in Microengineering

Core focus: mechanics, electronics, control systems, and manufacturing

Lausanne, Switzerland

Sep 2019 – Jul 2022

Selected Projects

Neuromuscular Adaptation to Exoskeleton Assistance

EPFL & UTS | 2025–2026

Developed a multimodal research framework combining EMG signal processing, Vicon motion capture, and biomechanical modeling to quantify adaptation under upper-limb exoskeleton assistance.

Applied muscle synergy analysis (NNMF) and biomechanical modeling to characterize neuromuscular adaptation mechanisms during robotic rehabilitation within an international collaboration between EPFL BioRob and the UTS Robotics Institute.

Rehabilitation Game & Instrumented Soles

Autonomy (Intern) | 2024–2025

Engineered a rehabilitation platform integrating custom pressure-sensing insoles, ESP32-based wireless telemetry, Bluetooth Low Energy communication, and a Unity biofeedback application.

Contributed across hardware design, embedded software, and system integration to support real-time gait monitoring and rehabilitation.

Autonomous Drone Navigation (Crazyflie)

EPFL | 2025

Designed and implemented a simulation-to-real autonomous flight pipeline using OpenCV-based gate detection, state estimation, and cascaded PID control.

Successfully deployed the system on physical Crazyflie hardware, enabling autonomous gate traversal in real-world flight experiments.

Autonomous Navigation for a Wheeled Robot

EPFL | 2023

Developed a complete perception, planning, and control stack for autonomous navigation in cluttered environments.

Integrated computer vision localization, A* planning, Kalman filtering, and reactive obstacle avoidance within a closed-loop autonomous navigation system.

Technical Skills

Programming

Python • C • C++ • C# • MATLAB • Git

Robotics & Control

Model Predictive Control (MPC) • PID Control • State Estimation • Kalman Filtering • Sensor Fusion • Computer Vision • Motion Planning (A*, Dijkstra)

Simulation & Biomechanics

MuJoCo • Webots • OpenSim • SCONE • Unity • EMG Signal Processing • Vicon Motion Capture

Embedded Systems & Electronics

ESP32 • STM32 • Bluetooth Low Energy (BLE) • Sensors & Actuators • PCB Design (KiCad)

Mechanical Design

Fusion 360 • SolidWorks • CATIA • Onshape • Rapid Prototyping

Optimization

CasADi • Gurobi • YALMIP

Software Engineering

Terraform • React • TypeScript • SQLite • Kivy

Languages

French (Native) • Persian (Native) • English (Professional Working Proficiency) • German (Basic)

Leadership

General Coaching Program, EPFL

Coordinated 50+ coaches across 13 EPFL departments, supporting large-scale student mentoring and integration initiatives.

Microengineering Coaching, EPFL

Mentored first-year Microengineering students to support academic integration and success.

Interests

Strength Training & Calisthenics • Mountain Hiking (20+ km treks, 1000m+ elevation gain) • Literature & Philosophy (Camus, contemporary novels)